

```

import sys
print(sys.executable)

C:\Users\LENOVO\anaconda3\python.exe

%pip install mysql-connector-python

Collecting mysql-connector-python
  Downloading mysql_connector_python-9.7.0-py2.py3-none-any.whl.metadata (11 kB)
  Downloading mysql_connector_python-9.7.0-py2.py3-none-any.whl (480 kB)
    ----- 0.0/480.6 kB ? eta -:-:-
    ----- 0.0/480.6 kB ? eta -:-:-
    ----- 10.2/480.6 kB ? eta
-:-:-
    ----- 10.2/480.6 kB ? eta
-:-:-
    ----- 10.2/480.6 kB ? eta
-:-:-
    ----- 41.0/480.6 kB 281.8 kB/s
eta 0:00:02
    ----- 41.0/480.6 kB 281.8 kB/s
eta 0:00:02
    ----- 112.6/480.6 kB 504.4 kB/s
eta 0:00:01
    ----- 112.6/480.6 kB 504.4 kB/s
eta 0:00:01
    ----- 235.5/480.6 kB 722.1 kB/s
eta 0:00:01
    ----- 235.5/480.6 kB 722.1 kB/s
eta 0:00:01
    ----- 471.0/480.6 kB 1.1 MB/s
eta 0:00:01
    ----- 480.6/480.6 kB 1.0 MB/s
eta 0:00:00
Installing collected packages: mysql-connector-python
Successfully installed mysql-connector-python-9.7.0
Note: you may need to restart the kernel to use updated packages.

import mysql.connector

import mysql.connector
import pandas as pd
import matplotlib.pyplot as plt

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="Dhanush",
    database="sales_db"
)

```

```
print("Connected Successfully!")
```

Connected Successfully!

```
query = "SELECT * FROM sales"
```

```
df = pd.read_sql(query, conn)
```

df

```
C:\Users\LENOVO\AppData\Local\Temp\ipykernel_3736\1415883195.py:3:
UserWarning: pandas only supports SQLAlchemy connectable
(engine/connection) or database string URI or sqlite3 DBAPI2
connection. Other DBAPI2 objects are not tested. Please consider using
SQLAlchemy.
```

```
df = pd.read_sql(query, conn)
```

	id	product	quantity	price
0	1	Laptop	2	60000.0
1	2	Mouse	5	500.0
2	3	Keyboard	3	1200.0
3	4	Laptop	1	60000.0
4	5	Mouse	7	500.0
5	6	Keyboard	2	1200.0
6	7	Monitor	4	15000.0
7	8	Monitor	1	15000.0

```
query = """
SELECT
    product,
    SUM(quantity) AS Total_Quantity,
    SUM(quantity * price) AS Revenue
FROM sales
GROUP BY product;
"""
```

```
sales_summary = pd.read_sql(query, conn)
```

sales_summary

```
C:\Users\LENOVO\AppData\Local\Temp\ipykernel_3736\3267992217.py:10:
UserWarning: pandas only supports SQLAlchemy connectable
(engine/connection) or database string URI or sqlite3 DBAPI2
connection. Other DBAPI2 objects are not tested. Please consider using
SQLAlchemy.
```

```
sales_summary = pd.read_sql(query, conn)
```

	product	Total_Quantity	Revenue
0	Laptop	3.0	180000.0
1	Mouse	12.0	6000.0

2	Keyboard	5.0	6000.0
3	Monitor	5.0	75000.0

```
query = """
SELECT
    SUM(quantity) AS Total_Items_Sold,
    SUM(quantity * price) AS Total_Revenue
FROM sales;
"""
```

```
overall_summary = pd.read_sql(query, conn)
```

```
overall_summary
```

C:\Users\LENOVO\AppData\Local\Temp\ipykernel_3736\1368141490.py:8:
UserWarning: pandas only supports SQLAlchemy connectable
(engine/connection) or database string URI or sqlite3 DBAPI2
connection. Other DBAPI2 objects are not tested. Please consider using
SQLAlchemy.

```
overall_summary = pd.read_sql(query, conn)
```

	Total_Items_Sold	Total_Revenue
0	25.0	267000.0

```
print("Sales Summary")
```

```
print(sales_summary)
```

```
print("\nOverall Sales Summary")
```

```
print(overall_summary)
```

Sales Summary

	product	Total_Quantity	Revenue
0	Laptop	3.0	180000.0
1	Mouse	12.0	6000.0
2	Keyboard	5.0	6000.0
3	Monitor	5.0	75000.0

Overall Sales Summary

	Total_Items_Sold	Total_Revenue
0	25.0	267000.0

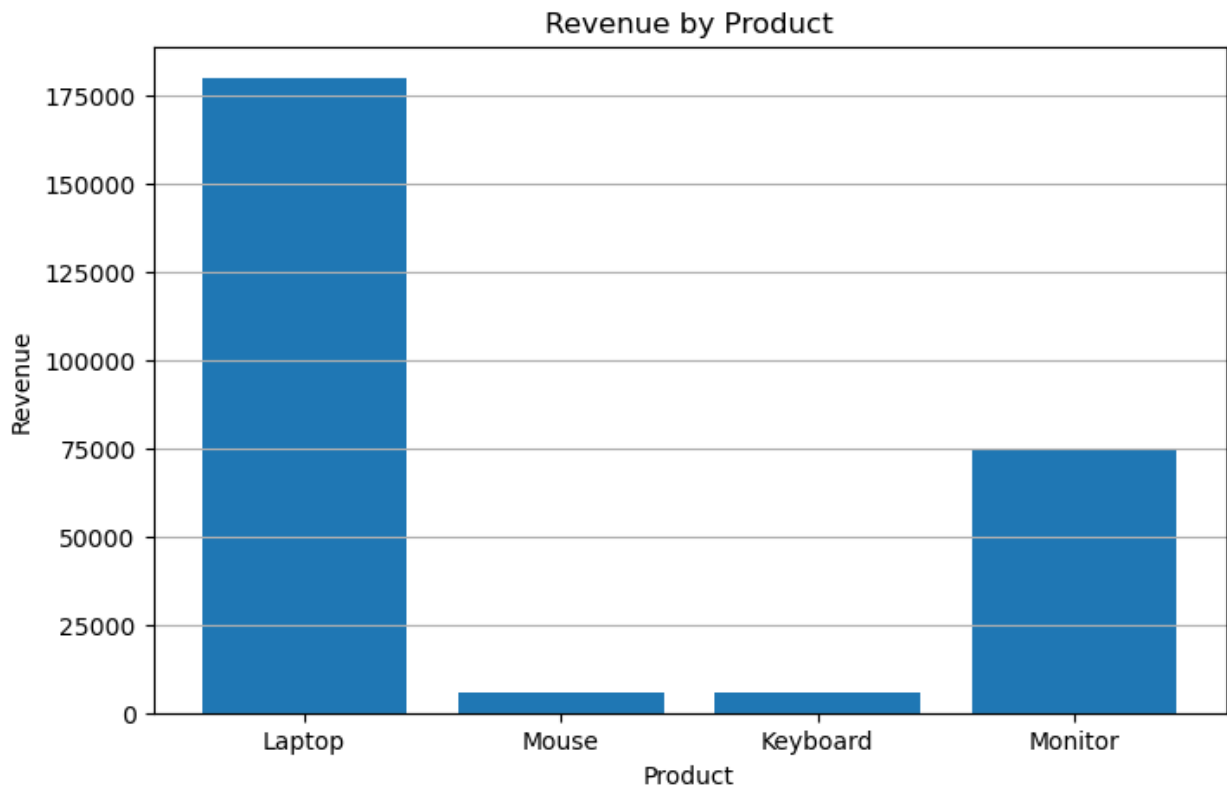
```
plt.figure(figsize=(8,5))
```

```
plt.bar(
    sales_summary["product"],
    sales_summary["Revenue"]
)
```

```
plt.title("Revenue by Product")
```

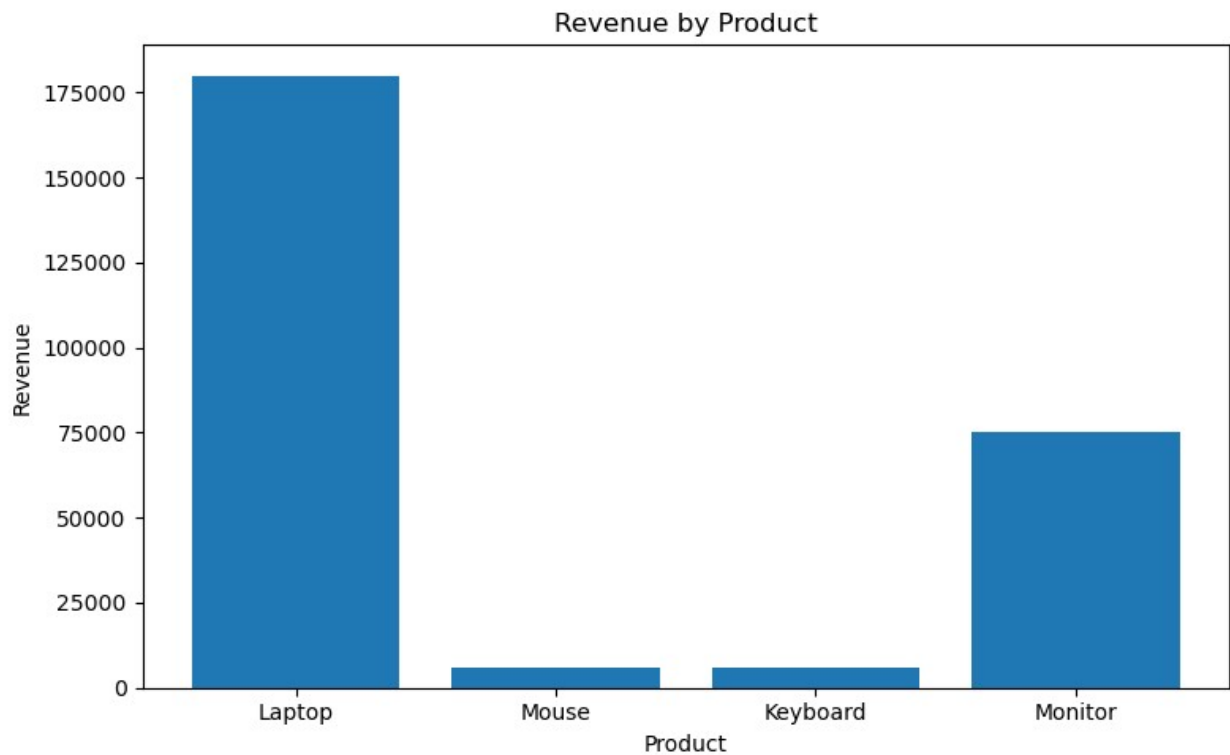
```
plt.xlabel("Product")
```

```
plt.ylabel("Revenue")
plt.grid(axis='y')
plt.show()
```



```
plt.figure(figsize=(8,5))
plt.bar(
    sales_summary["product"],
    sales_summary["Revenue"]
)
plt.title("Revenue by Product")
plt.xlabel("Product")
plt.ylabel("Revenue")
plt.tight_layout()
plt.savefig(
    r"C:\Users\LENOVO\Downloads\Sales_Project\charts\
Revenue_by_Product.png",
```

```
    dpi=300,  
    bbox_inches="tight"  
)  
plt.show()
```



```
query = """  
SELECT  
product,  
SUM(quantity*price) AS Revenue  
FROM sales  
GROUP BY product  
ORDER BY Revenue DESC  
LIMIT 1;  
"""  
  
highest = pd.read_sql(query, conn)  
  
print(highest)  
  
product  Revenue  
0  Laptop  180000.0
```

```
C:\Users\LENOVO\AppData\Local\Temp\ipykernel_3736\2314940762.py:11:
UserWarning: pandas only supports SQLAlchemy connectable
(engine/connection) or database string URI or sqlite3 DBAPI2
connection. Other DBAPI2 objects are not tested. Please consider using
SQLAlchemy.
```

```
highest = pd.read_sql(query, conn)
```

```
query = """
SELECT
product,
SUM(quantity) AS Quantity
FROM sales
GROUP BY product
ORDER BY Quantity DESC
LIMIT 1;
"""
```

```
highest_qty = pd.read_sql(query, conn)
```

```
print(highest_qty)
```

	product	Quantity
0	Mouse	12.0

```
C:\Users\LENOVO\AppData\Local\Temp\ipykernel_3736\2974785598.py:11:
UserWarning: pandas only supports SQLAlchemy connectable
(engine/connection) or database string URI or sqlite3 DBAPI2
connection. Other DBAPI2 objects are not tested. Please consider using
SQLAlchemy.
```

```
highest_qty = pd.read_sql(query, conn)
```

```
query = """
SELECT
product,
AVG(price) AS Average_Price
FROM sales
GROUP BY product;
"""
```

```
avg_price = pd.read_sql(query, conn)
```

```
print(avg_price)
```

	product	Average_Price
0	Laptop	60000.0
1	Mouse	500.0
2	Keyboard	1200.0
3	Monitor	15000.0

```
C:\Users\LENOVO\AppData\Local\Temp\ipykernel_3736\698404328.py:9:
UserWarning: pandas only supports SQLAlchemy connectable
```

(engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.

```
avg_price = pd.read_sql(query, conn)
```

```
query = """
SELECT
product,
COUNT(*) AS Number_of_Sales
FROM sales
GROUP BY product;
"""
```

```
sales_count = pd.read_sql(query, conn)
```

```
print(sales_count)
```

	product	Number_of_Sales
0	Laptop	2
1	Mouse	2
2	Keyboard	2
3	Monitor	2

C:\Users\LENOVO\AppData\Local\Temp\ipykernel_3736\1471481538.py:9:

UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.

```
sales_count = pd.read_sql(query, conn)
```

```
plt.figure(figsize=(8,5))
```

```
plt.bar(
    sales_summary["product"],
    sales_summary["Total_Quantity"],
    color="green"
)
```

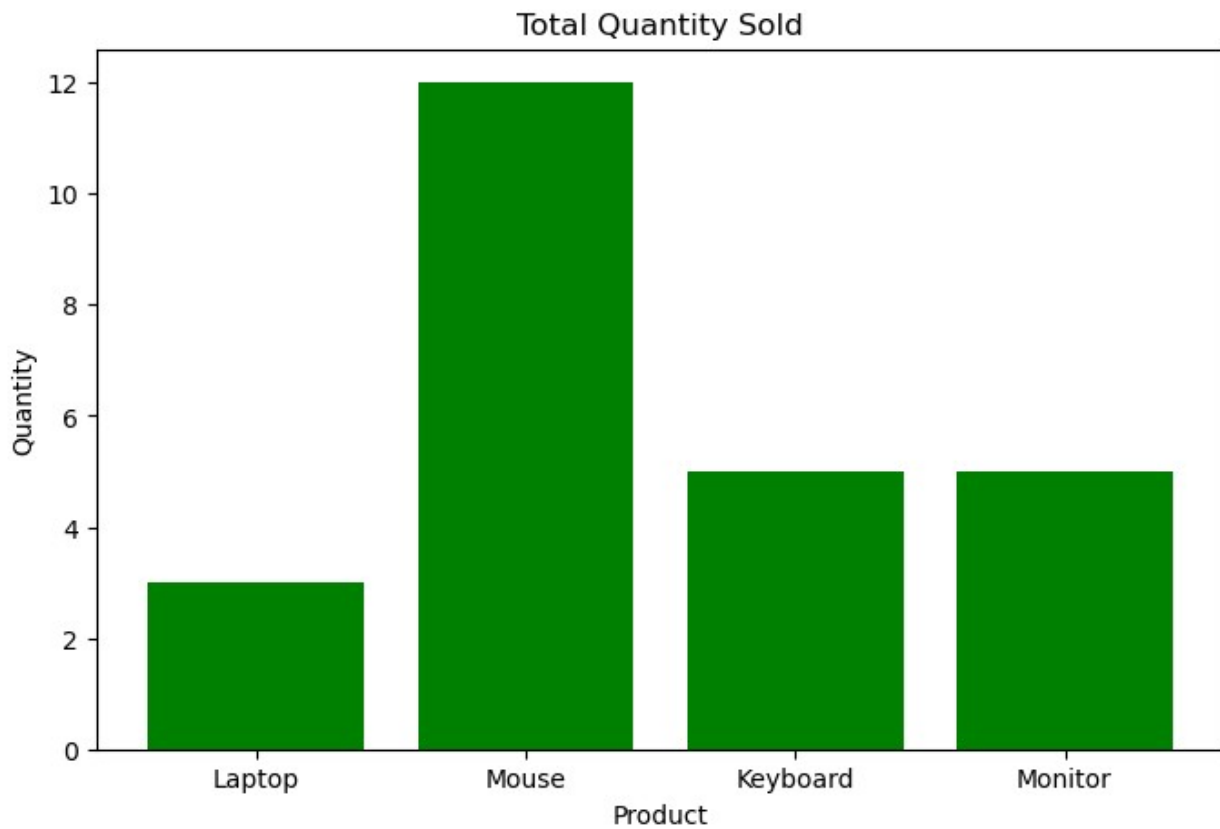
```
plt.title("Total Quantity Sold")
```

```
plt.xlabel("Product")
```

```
plt.ylabel("Quantity")
```

```
plt.savefig(
    r"C:\Users\LENOVO\Downloads\Sales_Project\charts\
Total_Quantity_Sold.png",
    dpi=300,
    bbox_inches="tight"
)
```

```
plt.show()
```



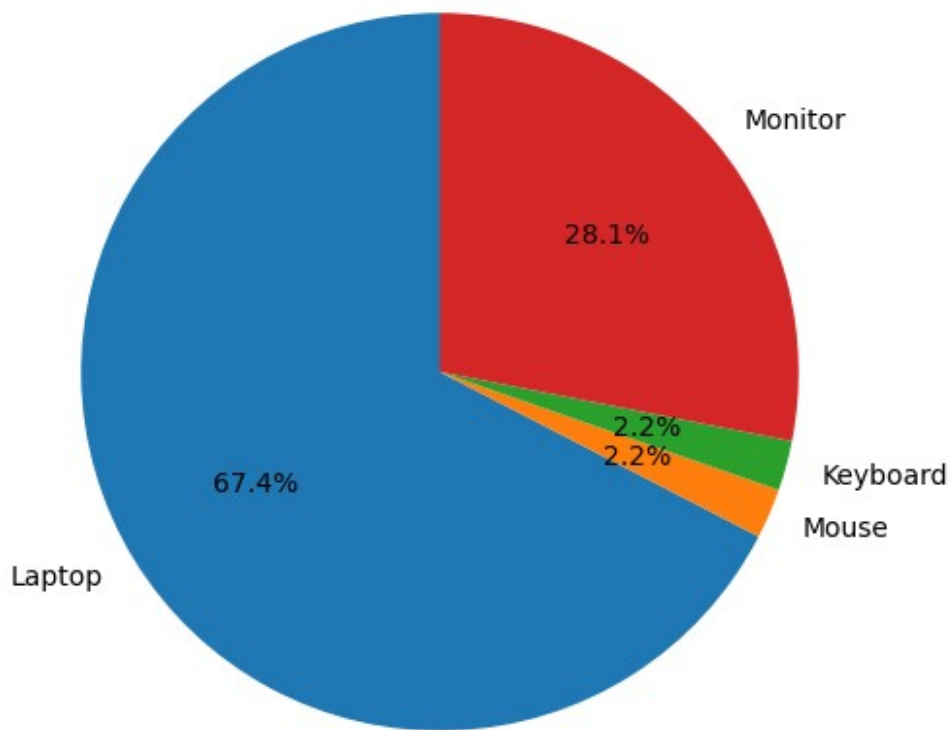
```
plt.figure(figsize=(6,6))

plt.pie(
    sales_summary["Revenue"],
    labels=sales_summary["product"],
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Revenue Distribution")
plt.savefig(
    r"C:\Users\LENOVO\Downloads\Sales_Project\charts\
Revenue_Distribution.png",
    dpi=300,
    bbox_inches="tight"
)

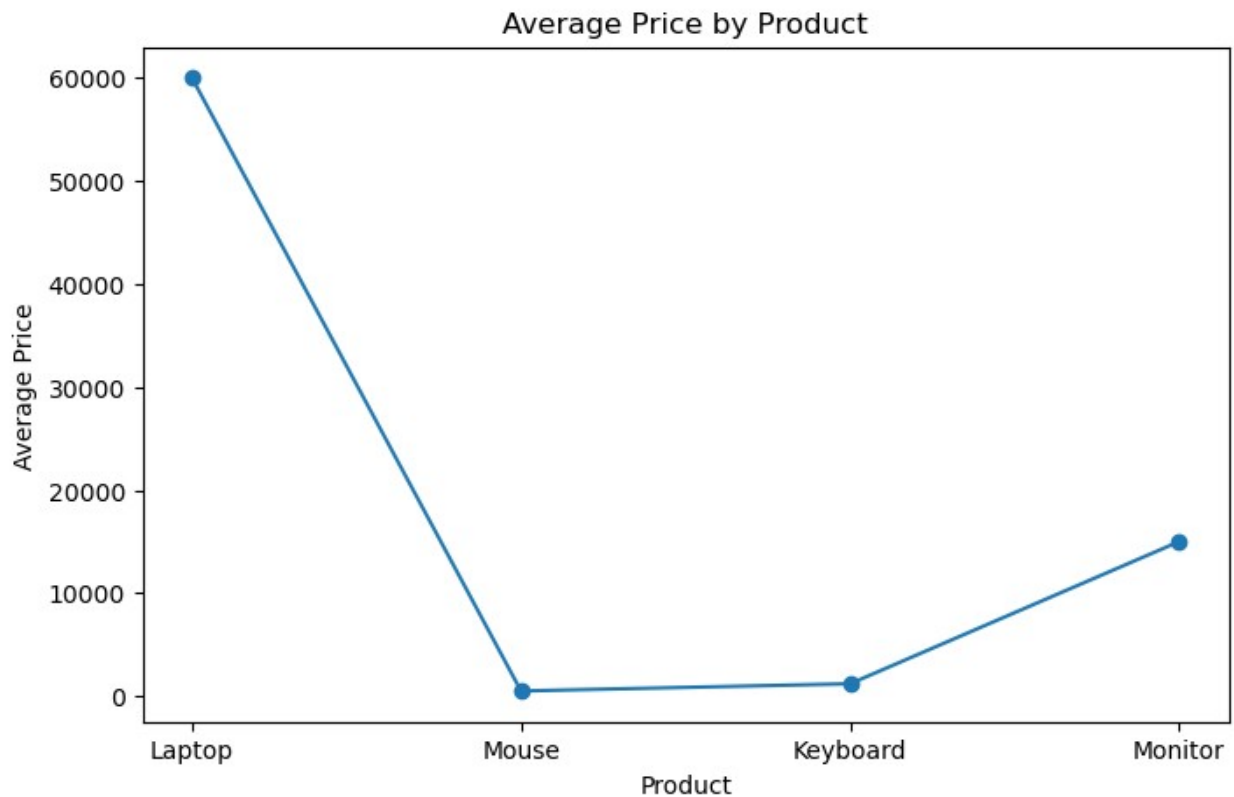
plt.show()
```


Revenue Distribution



```
plt.figure(figsize=(8,5))
plt.plot(
    avg_price["product"],
    avg_price["Average_Price"],
    marker="o"
)
plt.title("Average Price by Product")
plt.xlabel("Product")
plt.ylabel("Average Price")
plt.savefig(
    r"C:\Users\LENOVO\Downloads\Sales_Project\charts\
Average_Price.png",
    dpi=300,
    bbox_inches="tight"
)
```

```
plt.show()
```



```
query = """
SELECT
product,
SUM(quantity*price) AS Revenue
FROM sales
GROUP BY product
ORDER BY Revenue DESC;
"""

ranking = pd.read_sql(query, conn)

print(ranking)
```

	product	Revenue
0	Laptop	180000.0
1	Monitor	75000.0
2	Mouse	6000.0
3	Keyboard	6000.0

```
C:\Users\LEN0V0\AppData\Local\Temp\ipykernel_3736\2150777145.py:10:
UserWarning: pandas only supports SQLAlchemy connectable
(engine/connection) or database string URI or sqlite3 DBAPI2
```

connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.

```
ranking = pd.read_sql(query, conn)
```

```
import os
```

```
save_path = r"C:\Users\LENOVO\Downloads\Sales_Project\charts"
```

```
os.makedirs(save_path, exist_ok=True)
```

```
print("Folder Ready!")
```

```
Folder Ready!
```